## 5.6 Infinite Sequences and Series

A finite sum of real numbers is well defined by the properties of real numbers, but in order to make sense of an infinite series, we need to consider the concept of convergence. Consider the infinite sum: $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots$ with each term positive. Can we assign a numerical value to the sum? While at first it may seem difficult or impossible. We can certainly do something similar where we have one quantity getting closer and closer to a fixed quantity.

Let us discuss an interesting problem. Let there be two plates $A$ and $B$. Let a full cake be placed on the plate $A$ and let $B$ as empty. Let us cut the cake in $A$ into exactly two equal parts and place one part on $B$ leaving the other part in $A$. Let us cut the remaining part of the cake in $A$ into exactly two equal parts and place one part on $B$ leaving the other part in $A$. Let us again cut the remaining part of the cake in $A$ into exactly two equal parts and place one part on $B$ leaving the other part in $A$. If we go on doing this what will happen? What will be the amount of cake "*finally*" in $A$ and in $B$? Let us list the stage by stage status:

| Stage | Plate $A$ | Plate $B$ |
|---|---|---|
| 0 | 1 | 0 |
| 1 | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 2 | $\frac{1}{4}$ | $\frac{1}{2}+\frac{1}{4}$ |
| 3 | $\frac{1}{8}$ | $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}$ |
| 4 | $\frac{1}{16}$ | $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}$ |
| 5 | $\frac{1}{32}$ | $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}$ |
| ... | ... | ... |
| $n$ | $\frac{1}{2^n}$ | $\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\cdots+\frac{1}{2^n}$ |
| ... | ... | ... |

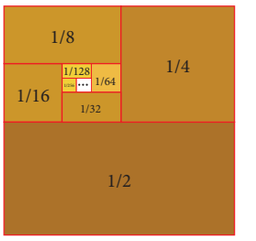



**Figure 5.3**

Intuitively we feel that "*finally*" nothing will remain in plate $A$ and the full cake will be in plate $B$. In other words, the cake available in $A$ is 0 and the cake available in $B$ is 1. That is, intuitively we feel that

$$1,\ \frac{1}{2},\ \frac{1}{4},\ \frac{1}{8},\ \frac{1}{16},\ \frac{1}{32},\ \cdots$$

"*goes*" to 0 and

$$\frac{1}{2},\ \frac{1}{2}+\frac{1}{4},\ \frac{1}{2}+\frac{1}{4}+\frac{1}{8},\ \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16},\ \cdots$$

"*goes*" to 1 or equivalently

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots \qquad \text{is 1.}$$

In this section let us learn the sense in which the words "*finally*" and "*goes*" are used and also let us learn the addition of infinitely many numbers.

We intuitively feel that $1,\ \frac{1}{2},\ \frac{1}{4},\ \frac{1}{8},\ \frac{1}{16},\ \frac{1}{32},\ \ldots$ "*goes*" to 0. Similarly we feel that the sequence $1,\ \frac{1}{10},\ \frac{1}{100},\ \frac{1}{1000},\ \frac{1}{10000},\ \frac{1}{100000},\ \ldots$ also "*goes*" to 0.

If $(a_n)$ is a sequence and $a$ is a number so that for any given small positive number, there is a stage after which the distance between $a_n$ and $a$ is smaller than that positive number, then we may say that $a_n$ goes to $a$ as $n$ goes to infinity. In technical terms we may say that $a_n$ *tends to* $a$ *as* $n$ *tends to infinity.* In other words, in the limiting case $a_n$ becomes $a$ or the limit of $a_n$ is $a$ as $n$ tends to $\infty$. We also say that the sequence $(a_n)$ converges to $a$. If $(a_n)$ converges to $a$, then we write $\lim\limits_{n\to\infty} a_n = a$.